

Other Continuous Distributions

Reading Objectives:

In this reading, you will understand the other continuous distributions, such as Uniform, Gamma, Exponential, and Beta distributions.

Main Reading Section:

Uniform Distribution

A random variable X is said to be uniformly distributed over the interval $[\alpha, \beta]$ if its probability density function is given by

$$f(x) = \begin{cases} \frac{1}{\beta-\alpha}, & \text{if } \alpha \leq x \leq \beta \\ 0, & \text{Otherwise} \end{cases}$$

Note that the foregoing meets the requirements of being a probability density function since

$$\frac{1}{\beta-\alpha} \int_{\alpha}^{\beta} dx = 1$$

The uniform distribution arises in practice when we suppose a certain random variable is equally likely to be near any value in the interval $[\alpha, \beta]$.

The probability that X lies in any subinterval of $[\alpha, \beta]$ is equal to the length of that subinterval divided by the length of the interval $[\alpha, \beta]$. This follows since when $[a, b]$ is a subinterval of $[\alpha, \beta]$

$$P\{a < X < b\} = \frac{1}{\beta-\alpha} \int_a^b dx = \frac{b-a}{\beta-\alpha}$$

Properties of Uniform Distribution

$$E[X] = \frac{\alpha+\beta}{2} \text{ and } \text{Var}(X) = \frac{(\beta-\alpha)^2}{12}$$

Gamma Distribution

A random variable is said to have a gamma distribution with parameters (α, λ) , $\lambda > 0$, $\alpha > 0$ if its density function is given by

$$f(x) = \begin{cases} \frac{\lambda e^{-\lambda x} (\lambda x)^{\alpha-1}}{\Gamma(\alpha)}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$$\begin{aligned} \text{Where } \Gamma(\alpha) &= \int_0^{\infty} \lambda e^{-\lambda x} (\lambda x)^{\alpha-1} dx \\ &= \int_0^{\infty} e^{-y} (y)^{\alpha-1} dy \end{aligned}$$

(by letting $y = \lambda x$), integration yields

$$= (\alpha - 1) \int_0^{\infty} e^{-y} y^{\alpha-2} dy$$

which gives

$$\Gamma(\alpha) = (\alpha-1) \Gamma(\alpha-1)$$

Properties of Gamma distribution

$$\Gamma(n) = (n-1)!$$

The moment generating function of a gamma random variable X with parameters (α, λ) is obtained as follows $\Phi(t) = E[e^{tX}] = \left(\frac{\lambda}{\lambda-t}\right)^{\alpha}$

$$E[X] = \Phi'(0) = \frac{\alpha}{\lambda} \text{ and } \text{Var}(X) = E[X^2] - E[X]^2 = \phi''(0) - \left(\frac{\alpha}{\lambda}\right)^2 = \frac{\alpha}{\lambda^2}$$

Exponential Distribution

A continuous random variable whose probability density function is given, for some $\lambda > 0$, by

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & \text{if } x \geq 0 \\ 0, & x < 0 \end{cases}$$

is said to be an exponential random variable (or, more simply, is said to be exponentially distributed) with parameter λ . The cumulative distribution function $F(x)$ of an exponential random variable is given by

$$F(x) = P\{X \leq x\} = \int_0^x \lambda e^{-\lambda y} dy = 1 - e^{-\lambda x}$$

Properties of exponential distribution

The moment generating function of a gamma random variable X with parameters (α, λ) is obtained as

$$\text{follows } \Phi(t) = E[e^{tX}] = \frac{\lambda}{\lambda - t}$$

$$E[X] = \Phi'(0) = \frac{1}{\lambda} \text{ and } \text{Var}(X) = E[X^2] - E[X]^2 = \Phi''(0) - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2}$$

Beta Distribution

The probability density function of the beta distribution, for $0 \leq x \leq 1$, and shape parameters $\alpha, \beta > 0$, is a power function of the variable x and its reflection $(1 - x)$ as follows:

$$f(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}$$

$$\text{where } B(\alpha, \beta) = \int_0^1 u^{\alpha-1} (1-u)^{\beta-1} du = \frac{\Gamma(\alpha) \Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

Properties of Beta distribution

$$E[X] = \frac{\alpha}{\alpha+\beta} \text{ and } \text{Var}(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$